ADITYA RAJ BHUSHAN

2nd year undergraduate
Institute of Engineering and Management, Newtown
Kolkata | West Bengal | Kolkata

+91 8240349445
bhushanraj1945@gmail.com
linkedin.com/in/adityarajbhushan

ACADEMIC PROFILE

DEGREE	INSTITUTE	PERCENTAGE (CGPA out of 10)	GRADUATION YEAR
B.Tech	IEM NEWTOWN (UEMK) - KOLKATA	8.19	2028

SEMISTER	SEM I	SEM II	SEM III	SEM IV	SEM V	SEM VI	SEM VII	SEM VIII
SGPA	8.43	7.97	8.19	-	-	-	-	-

ICSE (X)	HOLY CROSS SCHOOL	84%	2022
ISC(XII)	HOLY CROSS SCHOOL	78%	2024

SKILLS

- **Programming Languages:** C, Python, Java.
- **Engineering Software:** MATLAB, Multisim Online Simulator.
- **Hardware & Frameworks:** Arduino IDE, L293D Motor Drivers.
- **Areas of Interest:** Embedded Systems, IoT.

PROJECTS

Obstacle Avoiding Robot Using Arduino:
- Developed an autonomous obstacle detection and avoidance system using Arduino and ultrasonic sensors with real-time decision-making logic.
- Implemented motor control algorithms for dynamic navigation, ensuring efficient path correction and collision avoidance.

Train-Worker LoRa Safety System:
- Designed a LoRa-based long-range wireless safety system using ESP32 (SX1276/78) for real-time bidirectional communication between train and workers.
- Enabled worker detection, alert generation (OLED, buzzer, LED), and fail-safe monitoring to enhance railway safety.

PlantGuard AI:
- Built an AI system using YOLO object detection to detect pests and plant diseases from crop images in real time.
- Implemented a live webcam-based web application using Flask and OpenCV for browser-based pest monitoring.

Advanced Smart Parking System (IoT-Based, Commercial Grade):
- Designed and implemented a scalable parking solution with real-time detection and automated entry management.
- Integrated ESP32 WiFi and web interface for live monitoring, optimized for practical deployment.

HONOURS AND ACHIEVEMENTS

- Prize Winner – Google Arcade (November) for successful completion of technical challenges and tasks.
- Winner: Hacksnippet Hackathon 4.0 in IOT Track.
- Winner: Runners up Robohackers Hackathon UEMK.

POSITION OF RESPONSIBILITY

- Member I RASHMONI Raspberry Project Group: Led the structure department of the robot (2nd Year).
- Team Leader | Syntax Squad: Led a 4 member team for Hackathon.
- Team Leader | ArduTechTeam: Led a 5-member team for first-year technical projects and competitions (1st Year).